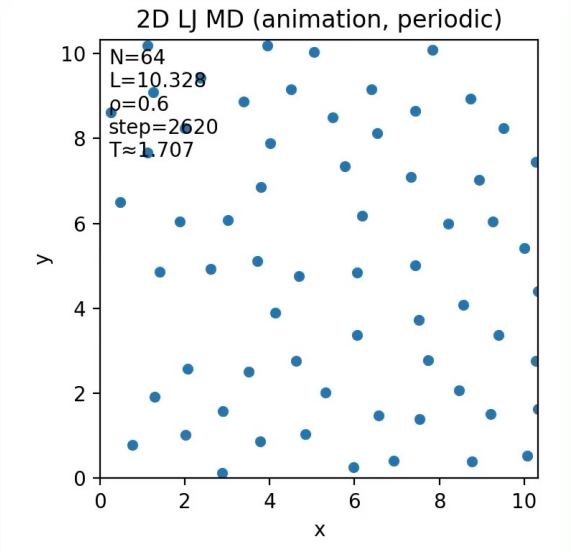
Monte Carlo Simulation

- **Model**: 2D Lennard-Jones potential
- **Result**: Molecular motion looked reasonable
- **Next goal**: Verify implementation correctness

  → Calculate argon vapor pressure curve

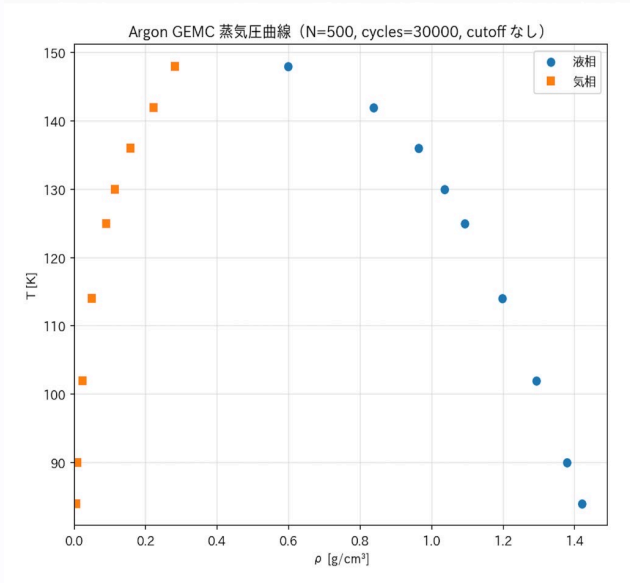
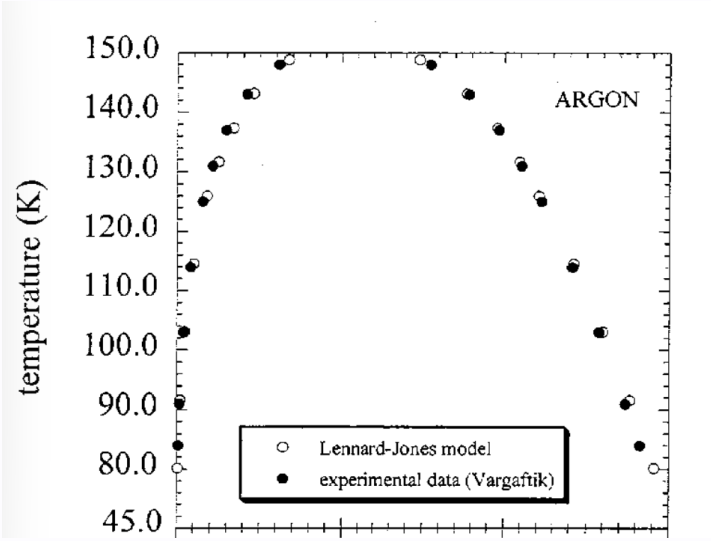


Argon Vapor Pressure Curve

- **GEMC** = Gibbs Ensemble MC (vapor–liquid coexistence)
- Python implementation(too slow) → migrated to Rust → speedup!

Computation time comparison (no cutoff, no cell list)

Condition	Python (s)	Rust (s)	Speedup
N=20, T=114K 40000 steps	273.3	0.30	~910×

Argon Vapor Pressure Curve

- Results agree well with prior studies